

STAT 218 Introductory Statistics and Data Analysis

Spring 2025

Lectures: Monday & Thursday at 9:55 am–11:10 am; Wednesday: 9:30 am–10:20 am

Classroom: Science Center L Wing 043

In-person lectures start on Wednesday, January 22, 2025. There are three lectures each week, including Wednesday, unless otherwise notified.

Instructor: Prof. Qing (Wendy) Wang

Email: qwang@wellesley.edu

Office: Science Center E452

Graders: Emily Suh and Zoe Zhou

Attached Tutor: Catherine Foster (cf104)

Office Hours and Tutoring Sessions:

- Monday: 1:30pm-2:30pm (SCI E452)
- Tuesday: 11:00am-12:00pm (Zoom, [link is here](#))
- Wednesday (tutor): 7:00-8:00pm (SCI H403)
- Thursday (tutor): 7:00-8:00pm (SCI H401)

Please note that the locations for the two weekly tutoring sessions are different.

The course attached tutor, Catherine Foster (cf104), is also available for one-on-one tutoring (30 minutes per week) throughout the semester. If you need additional help at any point, you may reach out to Catherine by email to schedule an individual meeting directly.

Important Dates:

January 22 (Wednesday): first class meeting

February 18 (Tuesday): Monday Schedule (we have class!)

March 17-21: no classes (Spring Break)

April 16 (Wednesday): no classes (Ruhlman Conference)

April 21 (Monday): no classes

May 1 (Monday Schedule Day): class cancelled, OHs held in my office during our class time.

Problem sets are due at 11:59pm on Thursdays, unless otherwise noted, and need to be submitted electronically through the Sakai course website.

Co/Prerequisite: MATH 205.

This course is not suited for students who are taking or have taken an introductory statistics course, such as ECON 103, PSYCH 105, STAT 160, or STAT/QR 260.

Textbook: Stats: Data and Models (5th edition), by De Veaux, Velleman, and Bock. Publisher: Pearson. Wellesley Bookstore link is [here](#).

One copy of the textbook (5th edition) is available on course reserve in the library (currently located in the Mods). Please refer to the correct edition of the textbook for assigned exercises in each problem set.

Course Description:

STAT 218 is an introductory statistics course. Topics covered include data collection, data visualization, descriptive statistics, regression, sampling, probability, statistical tests and inference, etc. Some statistical inferential tools and concepts are demonstrated through simulations, resampling, and permutation tests. A list of topics covered in STAT 218 can be found at the end of this syllabus. Previous knowledge in statistics is not required. Students should have taken **or** are taking MATH 205 (multivariable calculus). Students are expected to be familiar with partial derivatives and integration for up to two variables.

Distributions: Mathematical Modeling and Data Literacy

Learning Goals: After taking STAT 218, you should be able to

- understand basic and important concepts in statistics;
- present and summarize a data set with clear visualization and appropriate statistics;
- use linear regression models to study associations between variables;
- conduct hypothesis tests or construct confidence intervals to make statistical inference for an unknown population parameter of interest;
- program using statistical language R.

Computational Language:

R is an open source statistical programming language. The R language is the most used language among statisticians for data analysis, statistical simulations, and graphical display. R will be used throughout the semester for realizing data visualization, regression, statistical testing and inference. By the end of the semester, you will be very familiar with programming in R.

R can be downloaded from <http://www.cran.r-project.org/>. After R is installed on your own computer, you may consider to install RSTUDIO which is an integrated development environment (IDE) and provides a user-friendly interface. RSTUDIO is available from <http://www.rstudio.org/>.

Some reading materials and references for R can be found in the “R Help” folder on the course website on Sakai.

Grading Criteria:

Category	Percentage
Homework	25%
Lab Reports (3)	5%
Midterms (2)	(18+22)%
Final Exam	30%
Total	100%

- **Homework:**

The homework is worth 25% of the semester grade and consists of 9 assignments. Homework is due at 11:59pm on Thursdays, unless otherwise noted, and need to be submitted online via *Sakai's Assignment* tab. Each problem set is graded on a 100-point scale. The ***lowest one (1)*** grade will be dropped. No late submission is accepted without permission from the instructor at least 24 hours in advance of the due date.

Late Submission Policy:

- In general, late submission is not accepted unless a student has requested an extension at least 24 hours in advance.
- There is a one-time “forgiveness” policy. If a student accidentally misses a deadline by 30 minutes or less, their homework is accepted with a late penalty of 5 points off; submitting a homework 30 minutes to 1 hour late will incur 10 points deduction. No problem set is accepted 1 hour after the deadline, unless an extension has been approved in advance.

You are encouraged to discuss homework questions with others, but you must submit your own work. If R is used for some homework questions, you need to show your R commands and relevant output and include necessary comments. *One convenient way to prepare your solution for R-related questions is by using R MARKDOWN. A simple template and simple instructions on how to use R MARKDOWN can be found on Sakai. Using R Markdown for the problem sets is encouraged but not required.*

When submitting your completed assignment electronically online, it's best to combine all the pages into one single PDF document. Some recommendations and tips on how to create a PDF document can be found at the end of the syllabus.

Acceptable/Unacceptable use of AI:

- Generative AI presents a powerful tool that can significantly enhance one's learning experience. There are numerous effective ways to leverage generative AI to facilitate learning, such as requesting examples of R functions, or seeking clarification on complex statistical concepts. However, it is crucial to recognize the limitations of AI and avoid over-reliance on it. Below are occasions where the use of AI is not acceptable in this course.
- You should never upload a dataset from our homework into generative AI, or direct generative AI to use a particular dataset from any source.
- You should not copy your R code into generative AI, for debugging or any other reason.
- You should not copy homework question into generative AI.
- It is not acceptable to submit either written text or R code that you've copied from the output of generative AI, even if edited.

- **Lab Reports:**

There are three in-class R activities during the semester. You need to complete a lab report after each activity. The lab reports are marked for completion, not accuracy.

- **Exams:**

There are two midterm exams during the semester. The midterm exam has two parts: an in-class test (80%) and a take-home case study (20%). You are allowed to use a calculator and one page, single-sided of notes during the in-class test. Besides these, no other aid or materials are allowed. The take-home case study involves real data analysis using R and is assigned right after an in-class exam has taken place. The first in-class exam is tentatively scheduled on **March 6** (Thursday), and the second one is on **April 17** (Thursday). *The higher grade (out of the two midterms) will weigh 22%, while the lower grade accounts for 18%, when computing the overall performance in this course.*

The Final Exam is a fixed-schedule, cumulative written exam that takes place on a specific date during the final week. There is no data analysis portion for the final exam. You are allowed to bring *one-page, double-sided* of notes and a simple calculator to the test. No other aid or materials are allowed.

One week before each scheduled exam, a study guide and a practice exam will be posted on the course website to help you better prepare for the test.

Please note that graphical calculators are not allowed during exams, or you cannot use certain functions of your graphical calculators. For instance, you cannot use your calculator to compute simple calculus integrations or derivatives.

Conflicts on exam dates must be resolved in advance. An unexcused absence on an exam date may result in a score of 0 in the test.

- The Wellesley College Honor Code applies to all exams and coursework in this class. Please refer to the Student Handbook for details.

In particular, students are not allowed to look at any kind of solutions until the assignment is turned in.

Disability accommodations: Any student with accessibility or disability accommodation should contact Rachael Torella (rt107), preferably during the first 2 weeks of class, so that she can review the request and notify me of the approved accommodations in a timely manner. More information is available at the Accessibility and Disability Services website.

If you have documentation from the Office of Accessibility and Disability Resources that you'd like to share with me or anticipate conflicts with the course due to religious observance or a Wellesley-sponsored activity (such as an athletic team), please let me know early in the semester so that arrangements can be made.

Announcement:

Syllabus, lecture notes, homework solutions, quizzes, and all course materials will be posted on **Sakai**. You are expected to check this site frequently.

Help with STAT 218:

- Do not hesitate to raise any questions in class or after class. You can always reach me by email or stop by my office during office hours.
- Take advantage of my office hours and the attached tutor's weekly help sessions.

- Do not be afraid of making mistakes when using R. We always learn to master a programming language through trials and errors.
- The student-version textbook has solutions for all odd-number exercises. If you need extra practice questions, you could work out some of the odd-number questions and then compare your answers with the solutions. (I would also be happy to help you with questions related to these exercises.)

How to create a single PDF document?

- One that works for all cases, is to use this free website: <https://smallpdf.com/jpg-to-pdf>. SmallPDF is reliable, fast, simple, and secure.
- For iOS (iPhone or iPad):
 1. Find and select the pictures in the Photos app.
 2. Then choose Save to Files, and choose a folder to save it.
 3. Then launch the Files app to find them.
 4. Tap the Select button at the top right corner, then choose the More button (three dots) at the bottom right corner to bring up the options at the lower section on your phone screen.
 5. Choose Create PDF. A PDF file with the same name will be created and saved to the same folder as the original photos in Files app on your iPhone.
- For Android: get the free Image to PDF Converter app, which has a number of steps, but is basically similar to the above.
- On a Mac, select all the images, then open them with Preview. Adjust the order on the left, then choose Print, then click PDF at the bottom left, then Save as PDF.
- On Windows 10, use File Explorer to find the images. Select them in the order you want them to appear in the file, then Right-click and choose Print, then choose Microsoft Print to PDF as the “printer”.

Tentative Coverage and Schedule

	Topics	Textbook Chapters
1	Overview of Statistics	Chapter 1
2	Displaying categorical data	Chapter 2.1 & Chapter 3.1-3.3
3	Displaying quantitative data	Chapter 2.2-2.5
4	Boxplots and comparing observations in different groups	Chapter 4
	Using resampling schemes to compare two groups of observations*	
5	Standard deviation as a ruler and the Normal model	Chapter 5
6	Scatterplots, association, and correlation	Chapter 6
7	Simple linear regression models	Chapter 7 & Chapter 8
	Brief introduction to multiple linear regression models	Chapter 9
	R activity: the unbiasedness of Least Squares solution	
	R activity: Box-Cox transformation*	
8	Sample Surveys	Chapter 10
9	Observational study and experiments	Chapter 11
10	Probability	Chapter 12&13
11	Random variables	Chapter 14
12	Counting (combination, permutation, discrete random variables)	Chapter 15
13	Sampling distribution and Inference for proportions	Chapter 16 & Chapter 18.1-18.3
14	Inference about means	Chapter 17 & Chapter 18.4-18.6
	Power, Type I and Type II errors	Chapter 19
	R activity: Using bootstrap to approximate and understand p -value*	
15	Comparing two proportions or two means	Chapter 20
(16)	Chi-square tests	Chapter 22

*: marked are topics that are beyond the textbook coverage but will be introduced in class. If time permits, Topic 16 will be covered during the last week.